

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T20 · Completing the Square

With Answers

Questions with full worked solutions

■ 5 pages

FAMILY MAP

Completing the Square

F01

Complete the square for a non-monic quadratic

2 Q | 3 marks

19 $ABCED$ is a five-sided shape.

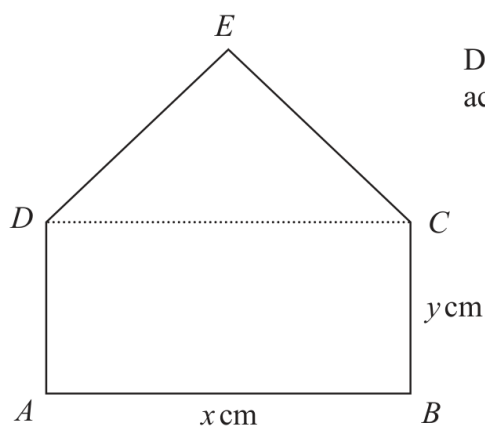


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

(1)

Worked solution

November 2021 | Question 19 | 2 marks

Family: Complete the square for a non-monic quadratic | Variant: Negative leading coefficient or maximum form

METHOD / WORKING

1. The perimeter is made from three sides of length x and two sides of length y :

2. $3x+2y=100$

3. So

4. $y = \frac{100-3x}{2}$

5. The area is the area of the rectangle plus the area of the equilateral triangle:

6. $R = xy + \frac{\sqrt{3}}{4}x^2$

7. Substitute $y = \frac{100-3x}{2}$:

8. $R = x(\frac{100-3x}{2}) + \frac{\sqrt{3}}{4}x^2$

9. $R = 50x - \frac{3}{2}x^2 + \frac{\sqrt{3}}{4}x^2$

10. $R = \frac{x}{4}(200 - (6 - \sqrt{3})x)$

11. For the maximum, write

12. $R = -\frac{6-\sqrt{3}}{4}x^2 + 50x$

13. The maximum occurs at

14. $x = \frac{-b}{2a} = \frac{-50}{2(-\frac{6-\sqrt{3}}{4})}$

15. $x = \frac{100}{6-\sqrt{3}}$

16. The coefficient of x^2 is negative, so the quadratic graph opens downwards and this vertex gives a maximum.17. Answer: $R = \frac{x}{4}(200 - (6 - \sqrt{3})x)$, and $x = \frac{100}{6-\sqrt{3}}$.

FINAL ANSWER

$$R = \frac{x}{4}(200 - (6 - \sqrt{3})x), \text{ and } x = \frac{100}{6-\sqrt{3}}$$

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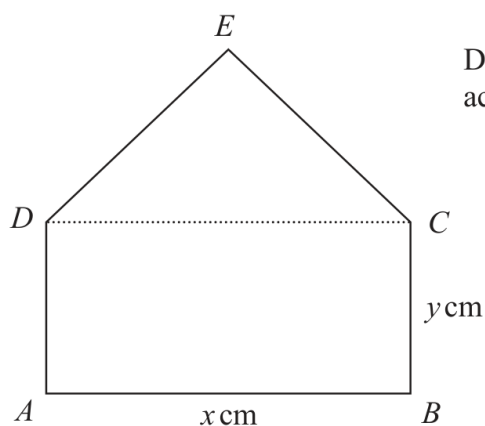


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Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

(1)

Worked solution

November 2021 | Question 19 | 1 marks

Family: Complete the square for a non-monic quadratic | Variant: Negative leading coefficient or maximum form

METHOD / WORKING

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